Dynamic Task Space Control of Redundant Pneumatically Actuated Soft Robot

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***Abstract*—Soft robotic manipulators promise manipulation dexterity and compliance, but these properties make them hard to control due to the uncertainties and nonlinearities in their dynamics. Towards solving these challenges, this letter proposes a real-time dynamic control approach for a pneumatically actuated redundant soft robotic arm. We first present a computationally-efficient dynamic model designed for real-time control. Next, we introduce a robust adaptive passivity-based control approach that complements the dynamic model to enhance the system's tracking performance under uncertainty. System uncertainties are categorized based on the bounds and behavior of dynamic terms in the model, with dominant stiffness and damping effects compensated by adaptive terms and unmodeled dynamics handled through the robust term. We first develop a control strategy in actuator space, then extend it to task space to encompass position control, orientation control, and simultaneous position and orientation control. To improve tracking performance in task space, we employ a redundancy resolution algorithm, addressing secondary tasks such as maintaining actuators within feasible limits and promoting homogeneous deformations in the robot. We comprehensively assess the performance of the proposed robust adaptive passivity control under various operation speeds, loading conditions, and trajectories. A comparison with recent baselines from the literature highlights how our method achieves both superior tracking performance and robot operation speed.**

***Index Terms*—Soft robot, dynamic, adaptive control, task space control.**

I. Introduction

IN MODERN robotics research, one of the fundamental goals is to replicate the remarkable abilities found in biological systems. This has led to growing investment in the field of soft robotics as researchers attempt to mimic the elastic and continuum structures discovered in animals [1]. Soft robots are unique due to their elasticity and ability to deform in their entirety. This allows them to navigate in confined spaces and adapt to various shapes while being intrinsically safe in human-robot interactions [2].

To fully realize the potential of soft robots, accurately modeling their behavior is essential. Cosserat rod modeling [3], [4] and finite element methods [5] capture the mechanical complexity and continuity of their structure with high accuracy for simulations and behavioral understanding. However, these methods are difficult to use for real-time control due to the computational complexity arising from their use of partial differential equations [1].

Simplifying assumptions can reduce model complexity while maintaining acceptable accuracy. A notable simplification of kinematic modeling for soft continuum robots is the piecewise constant curvature (PCC) assumption [6], which divides the continuous robot shape into finite and discrete constant curvature approximations. Combining the PCC assumption with a Lagrange formulation results in tractable ordinary differential equations of dynamic equation of motion. Della Santina et al. [7] proposed a dynamic model combining the PCC approach with an augmented rigid robot formulation. Furthermore, Godage et al. [8] assumed constant mass within each section of the PCC model while enabling variations in system volume and density. They introduced an integral formulation to capture the continuum effect of the system, which further increased model accuracy and applicability. However, the PCC assumption limits the accuracy of these models. To avoid the complexity of nonlinear terms that makes real-time use inefficient, researchers have instead explored complementary control schemes to address uncertainties and unmodeled dynamics in soft robotic systems [7], [9], [10].

A. Related Work

Here, we briefly review related model-based works focused on the dynamic control of soft robots. Della Santina et al. [7] used PD + feedback linearization to control a 5 degree-of-freedom (DoF) planar robot arm in joint space, with frequencies ranging from 0.1 rad/s to 10 rad/s. Falkenhahn et al. [11] implemented a similar algorithm for the dynamic control of a 9 DoF, 3D robot in actuator space with slowly varying set points. This approach was extended to task space for a 6 DoF manipulator, though limited to position control, by Fischer et al. [9], where they achieved a maximum tracking speed of 0.12 m/s and 0.012 m/s for pick-and-place tasks. The primary limitation of PD + feedback linearization is that it requires the exact cancellation of system dynamics, which is rarely achievable for soft robots due to their highly nonlinear elastic materials and uncertainties introduced by manufacturing inconsistencies and simplifying modeling assumptions.

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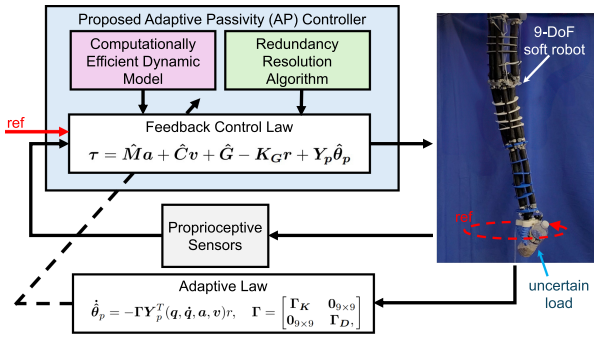


Fig. 1. Overview of our proposed approach on our soft arm.

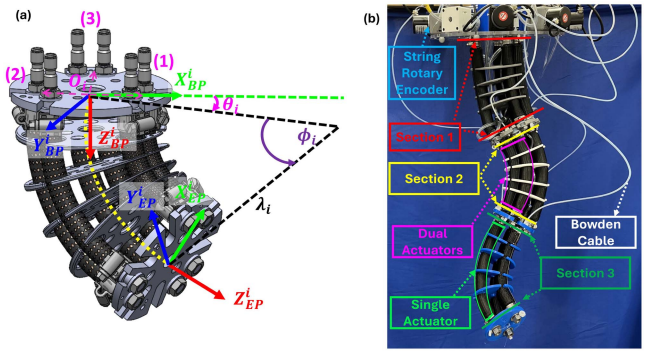


Fig. 2. (a) Notation for forward kinematics, illustrating configuration space variables and reference frames. "BP" and "EP" denote the base plate and end plate of section i, respectively. (b) The 9 DoF pneumatic soft arm consists of three sections, each with three actuation units. The actuation units in the first two sections have dual actuators, while those in the third section have single actuators.

Some recent work [12], [13] introduced an adaptive passivity control scheme that leveraged available model information and adapted to uncertainties for 3 DoF robot control in actuator space. This resulted in higher tracking accuracy and speed, setting it apart from earlier methods that used traditional PD + feedback linearization, though it was not demonstrated for task space control. Other research groups have also focused on addressing uncertainties. For instance, Kazemipour et al. [10] introduced an adaptive terminal sliding mode control for a 6 DoF, 3D arm, achieving a maximum trajectory speed of 0.11 m/s and an acceleration of 0.01 m/s^2, though only in position control. Bruder et al. [14], [15] attempted to identify the dynamics of a 6 DoF, 3D arm first using Koopman operator and then control the robot by model predictive methods but tracking was achieved only at quasi-static speeds.

B. Contributions of the Proposed Work

The primary objective of this work is to extend prior methods on adaptive passivity control [12], [13] from joint space to task space and implement the methods in a redundant 9 DoF pneumatically-actuated soft robotic arm, shown in Fig. 1. This was achieved via the following contributions: (i) exploiting structure in the dynamic model to improve computation speed, (ii) a redundancy resolution algorithm that ensures the robot obeys actuation limits while maintaining smooth motion by encouraging homogeneous deformation across the robot's sections, (iii) a demonstration for the first time of simultaneous position and orientation control despite the lack of a wrist joint, enabled by our redundancy resolution algorithm, and (iv) leveraging adaptive passivity control to address the uncertainties from the dynamic model that complements our acceleration level redundancy resolution algorithm.

To our knowledge, this is the first demonstration of dynamic task space control for a soft robot that simultaneously manages position and orientation while effectively avoiding actuator limits, implemented successfully on real hardware.

The remainder of the letter is structured as follows: Section II introduces our robot, Section III presents its kinematics and dynamics, Section IV outlines our proposed control framework, Section V presents experiments, and Section VI discusses our results and future research directions.

II. System Hardware Overview

We develop a modeling and control approach for a three-section pneumatic soft arm, shown in Fig. 2. Each section contains three actuation units evenly distributed around the center, and four 5 mm thick intermediate plates that enable the PCC constant curvature assumption. In the first and second sections, we use coupled *dual actuators* that are driven simultaneously within each actuation unit to increase stiffness and stability. In the distal section, we enable greater dexterity with *single actuators* in each actuation unit. To make the arm more compact, the second section is rotated by 60° relative to the first and third sections.

The system is actuated by nine proportional pressure valves, controlled by a Simulink xPC target machine. Actuator elongation is measured via custom-designed string rotary encoders, each consisting of a standard quadrature encoder attached to a 3D-printed spool mechanism and pre-loaded by a rotary spring. For each actuator, a monofilament string runs from the arm through a flexible Bowden cable sheath to the encoder spool. The sheaths ensure that shape changes in one section do not affect measurements in another. This design provides a linear resolution of approximately 0.03 mm, and the measured actuator lengths feed a forward kinematics model that determines the arm's tip pose. Control and measurements in the system run at a 1 kHz sampling rate. Further design information is detailed in our previous work [16].

III. Kinematic and Dynamic Modeling Review

In this section, we review the kinematic and dynamic modeling detailed in [8], [12], [13], [16] as applied to our three-section robot arm.

A. Kinematic Modeling

We assume that our robot follows the PCC model [13], [17]. For each section i, we employ mappings from actuator space to configuration space and then to task space.

1) Actuator Space to Configuration Space Mapping: This mapping translates elongation from the initial length L_0 of the three actuators, $\boldsymbol{q}_i = [l_{i,1}, l_{i,2}, l_{i,3}]^T$, into geometric shape changes, denoted as $\boldsymbol{\mathcal{C}}_i = [\lambda_i, \phi_i, \theta_i]^T$. Here, $\boldsymbol{q}_i$ represents the actuator space variables and $l_{i,j}$ denotes the elongation of each actuation unit j. Additionally, $\boldsymbol{\mathcal{C}}_i$ represents the configuration variables, where $\lambda_i \in (0, \infty)$ is the radius of curvature, $\phi_i \in [0, 2\pi)$ is the bending angle of the arc, and $\theta_i \in [-\pi, \pi]$ is the

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bending angle with respect to the $\mathbf{X}_i$ axis. The $\mathbf{X}_i$ axis is defined from the center of the base plate of section i toward the first actuation unit. The $\mathbf{Z}_i$ axis is perpendicular to the base plate and points toward the end plate. Finally, the $\mathbf{Y}_i$ axis is defined based on the right-hand rule (detailed derivation is presented in [13]).

2) Configuration Space to Task Space Mapping: Let $\mathbf{T}_i(\zeta_i, \boldsymbol{q}_i) \in \mathbb{SE}(3)$ be the task space frame of section i, where $\zeta_i \in [0, 1]$ indicates the placement of the moving frame within section i. When $\zeta_i = 0$, the moving frame is at the base plate and when $\zeta_i = 1$, the moving frame is at the end plate of the section. As in the literature [13], we use the PCC assumption to compute

$$\begin{aligned}\mathbf{T}_i(\zeta_i, \boldsymbol{q}_i) = \text{Rot}_z(\theta_i)\text{Trans}_x(\lambda_i)\text{Rot}_y(\zeta_i\phi_i) \\ \times \text{Trans}_x(-\lambda_i)\text{Rot}_z(-\theta_i)\end{aligned} \tag{1}$$

where $\text{Rot}_k \in \mathbb{SE}(3)$, and $\text{Trans}_k \in \mathbb{SE}(3)$ denote rotation and translation matrix along the k axis. From this, we derive frame position $\boldsymbol{P}(\zeta_i, \boldsymbol{q}_i) \in \mathbb{R}^3$ and rotation $\boldsymbol{R}(\zeta_i, \boldsymbol{q}_i) \in \mathbb{SO}(3)$ as

$$\mathbf{T}_i(\zeta_i, \boldsymbol{q}_i) = \begin{bmatrix} \mathbf{R}_i(\zeta_i, \boldsymbol{q}_i) & \boldsymbol{P}_i(\zeta_i, \boldsymbol{q}_i) \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix}. \tag{2}$$

3) Forward Kinematics of the Multi Section Soft Arm: Using forward kinematics in each section, we derive a homogeneous transformation matrix $\mathbf{T}^n$ of a moving frame along the robot's length as

$$\begin{aligned}\mathbf{T}^n(\boldsymbol{\zeta}^n, \boldsymbol{q}^n) &= \prod_{i=1}^{n}[\mathbf{T}_i(\zeta_i, \boldsymbol{q}_i)\text{Trans}_z(b_i)\text{Rot}_z(\beta_i)] \\ &= \begin{bmatrix} \boldsymbol{\Theta}_n(\boldsymbol{\zeta}^n, \boldsymbol{q}^n) & \boldsymbol{\Psi}_n(\boldsymbol{\zeta}^n, \boldsymbol{q}^n) \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix},\end{aligned} \tag{3}$$

where $\text{Trans}_z(b_i)$ and $\text{Rot}_z(\beta_i)$ are the translational and rotational offsets between plates i and $i+1$, $\boldsymbol{\Theta}_n \in \mathbb{SO}(3)$ and $\boldsymbol{\Psi}_n \in \mathbb{R}^3$ represent the rotation and position of the moving frame, and $\boldsymbol{\zeta}^n = [\zeta_1, \zeta_2, \ldots, \zeta_n]^T$, with $\zeta_i = 1$ for $i < n$ and $\zeta_i = 0$ for $i > n$. Additionally, $\boldsymbol{q}^n = [\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_n]$, and, unless stated otherwise, $\boldsymbol{q} = \boldsymbol{q}^3$, since $n = 3$ for our robot.

4) Differential Kinematics: Next, we derive the differential kinematics, or Jacobian, of the robot. To do this, we first review some notation. The hat operator $[\,\hat{}\,]$ maps $\mathbb{R}^3$ to $\mathbb{SO}(3)$, converting a 3D vector into its skew-symmetric matrix. The inverse, $[\,]^\vee$, maps from $\mathbb{SO}(3)$ back to $\mathbb{R}^3$, where $(\hat{a})^\vee = a$. Then, we calculate linear and angular velocities [8] as

$$\begin{aligned}\boldsymbol{v}_n(\boldsymbol{\zeta}^n, \boldsymbol{q}^n) &= \boldsymbol{\Theta}_n^T\dot{\Psi}_n = \mathbf{J}_n^v(\boldsymbol{\zeta}^n, \boldsymbol{q}^n)\dot{\boldsymbol{q}} \\ \boldsymbol{\omega}_n(\boldsymbol{\zeta}^n, \boldsymbol{q}^n) &= \left(\boldsymbol{\Theta}_n^T\dot{\boldsymbol{\Theta}}_n\right)^\vee = \mathbf{J}_n^\omega(\boldsymbol{\zeta}^n, \boldsymbol{q}^n)\dot{\boldsymbol{q}}\end{aligned} \tag{4}$$

where $\mathbf{J}_n^v \in \mathbb{R}^{3\times 3n}$, and $\mathbf{J}_n^\omega \in \mathbb{R}^{3\times 3n}$, are the linear and angular Jacobians in the body frame. These are calculated as

$$(\mathbf{J}_n^v)_k = \left(\boldsymbol{\Theta}_n^T\frac{\partial \boldsymbol{\Psi_n}}{\partial \boldsymbol{q(k)}}\right), \quad (\mathbf{J}_n^\omega)_k = \left(\boldsymbol{\Theta}_n^T\frac{\partial \boldsymbol{\Theta_n}}{\partial \boldsymbol{q(k)}}\right)^\vee \tag{5}$$

where $()_k$ denotes the kth column in the matrix, which is associated with actuator space variable $q(k)$.

B. Dynamic Modeling

We derive the system's closed form equations of motion (EoM), which are valuable for both control design and investigating system properties [18], using Lagrange's method:

$$\boldsymbol{M(q)\ddot{q}} + \boldsymbol{C(q, \dot{q})\dot{q}} + \boldsymbol{Kq} + \boldsymbol{D\dot{q}} + \boldsymbol{G(q)} = \boldsymbol{J}^T\mathbf{F}_{\boldsymbol{ext}} + \boldsymbol{\tau} \tag{6}$$

where $\boldsymbol{M} \in \mathbb{R}^{9\times 9}$ is the inertia matrix, $\boldsymbol{C} \in \mathbb{R}^{9\times 9}$ is the Centrifugal-Coriolis matrix, $\boldsymbol{K} \in \mathbb{R}^{9\times 9}$ is the stiffness coefficient matrix, $\boldsymbol{D} \in \mathbb{R}^{9\times 9}$ is the damping coefficient matrix, $\boldsymbol{G} \in \mathbb{R}^9$ is the gravity vector, $\mathbf{F}_{ext}$ is the external load, $\boldsymbol{J} = [\boldsymbol{J}_v^T, \boldsymbol{J}_\omega^T]^T$ is the Jacobian of where the external force $(\mathbf{F}_{\boldsymbol{ext}})$ is applied, and $\boldsymbol{\tau} \in \mathbb{R}^9$ is the input torque.

1) Inertia Matrix: The inertia matrix is derived as

$$\boldsymbol{M} = \boldsymbol{M}^v + \boldsymbol{M}^\omega \tag{7}$$

where $\boldsymbol{M}^v$ and $\boldsymbol{M}^\omega$ are the generalized linear and angular inertia matrices, respectively, calculated as

$$\mathbf{M}(\boldsymbol{q}) = \sum_{i=1}^{3}\int_{\zeta^i}\left\{(\mathbf{J}_i^v)^T\mathcal{M}_i^v\mathbf{J}_i^v + (\mathbf{J}_i^\omega)^T\mathcal{M}_i^\omega\mathbf{J}_i^\omega\right\}. \tag{8}$$

This integration is performed by assuming that the mass of each section is constant, while the density varies uniformly along its length [8]. Therefore, we can introduce a moving disk that sweeps along the length of section i to represent its mass element as $\delta m_i = m_i\delta\zeta_i$. This yields a linear inertia matrix, $\mathcal{M}_i^v = m_i\delta\zeta_i\mathbf{I}_3$, and an angular inertia matrix $\mathcal{M}_i^\omega = I_{xx}\delta\zeta_i\text{diag}(1, 1, 2)$.

2) Centrifugal-Coriolis Matrix: Per [18], we calculate the Centrifugal-Coriolis matrix as

$$C[k_3, k_2] = \sum_{k_1=1}^{9}\Gamma_{[k_1,k_2,k_3]}(M)\dot{q}(k_1) \tag{9}$$

where $\Gamma_{[k_1,k_2,k_3]}$ is Christoffel's symbol and $k_1, k_2, k_3 \in \{1, 2, \ldots, 9\}$ are placeholders for actuator variables indices.

3) Gravity Vector: Considering the gravitational acceleration vector $\boldsymbol{g} = [0, 0, g]^T$ and previous kinematic parameters, we can calculate the gravity vector as

$$\boldsymbol{G} = \sum_{i=1}^{3}\boldsymbol{G}_i, \quad [\boldsymbol{G}_i]_k = m_i\int_{\zeta^i}(\mathbf{J}_i^v)_k^T\boldsymbol{\Theta}_i^T\boldsymbol{g} \tag{10}$$

4) Stiffness and Damping Matrix: The stiffness and damping of each actuator have been shown to be functions of pressure and deformation [13]. However, in this study, we consider the stiffness and damping coefficients K and D with nominal parts K_n and D_n, and uncertainties ΔK_n and ΔD_n, such that $K = K_n + \Delta K_n$ and $D = D_n + \Delta D_n$. We address these uncertainties using a closed-loop robust adaptive control scheme. In the modeling phase, we treat the stiffness and damping coefficients as constants, represented by $\mathbf{K} \in \mathbb{R}^{9\times 9}$ and $\mathbf{D} \in \mathbb{R}^{9\times 9}$, where $\mathbf{K}$ and $\mathbf{D}$ are diagonal matrices of the stiffness and damping coefficients of all the actuators, respectively. Specifically, $\mathbf{K} = \text{diag}(K_1, K_2, \ldots, K_9)$ and $\mathbf{D} = \text{diag}(D_1, D_2, \ldots, D_9)$.

C. Taylor Series Approximation and Recursive Calculation

When employing the PCC assumption, numerical singularity issues occur when the sections become straight, leading

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to intractable calculations for kinematic and dynamic terms. To resolve this problem, we use Taylor series expansion to approximate the kinematic definitions, which also modifies the calculation of dynamic terms. Based on the findings of [8], we adopt an 11th-order approximation, which provides satisfactory accuracy for both kinematic and dynamic terms. Furthermore, to enhance the efficiency of these calculations, we implement the recursive formulation introduced in [8].

IV. Proposed Control Scheme

In this section, we propose a control strategy for a 9 DoF pneumatically actuated soft arm. Our primary focus is twofold: (i) developing a computationally efficient, real-time control method that manages the arm's complex dynamics, and (ii) formulating the control in both actuator and task spaces to achieve position, orientation, and simultaneous position and orientation control, while respecting actuator-limit constraints.

A. Modeling Assumptions

To achieve these objectives, we make the following assumptions: (AS-i) The inertia matrix, $\boldsymbol{M}$ in (7), can be approximated by $\boldsymbol{M}^v$, because the rotational kinetic energy is negligible compared to the translational kinetic energy [11]. (AS-ii) The Coriolis and Centrifugal matrix, $\boldsymbol{C}\dot{\boldsymbol{q}}$, can be neglected due to computational intensity and its minimal effect in soft robotics [11]. (AS-iii) The stiffness and damping coefficients are functions of input pressure and elongation, and vary slowly over time [19]. (AS-iv) The elastic behavior of soft robots is the dominant factor in this system [1].

B. Actuator Space Control

The goal of actuator space control is to track a predefined actuator space trajectory. We assume that a trajectory optimization algorithm has generated a desired trajectory (i.e., acceleration and velocity) for actuator variables from an intended task space trajectory. Then, we seek to enable the robot to achieve the desired accelerations and velocities. To do so, first, we define auxiliary error signals

$$\boldsymbol{v} = \dot{\boldsymbol{q}}_d - \boldsymbol{\Lambda}\tilde{\boldsymbol{q}}, \quad \boldsymbol{a} = \dot{\boldsymbol{v}} = \ddot{\boldsymbol{q}}_d - \boldsymbol{\Lambda}\dot{\tilde{\boldsymbol{q}}}, \quad \boldsymbol{r} = \dot{\boldsymbol{q}} - \boldsymbol{v} = \dot{\tilde{\boldsymbol{q}}} + \boldsymbol{\Lambda}\tilde{\boldsymbol{q}} \tag{11}$$

where $\boldsymbol{\Lambda} \in \mathbb{R}^{9\times 9}$ is a positive definite diagonal matrix and $\tilde{\boldsymbol{q}} = \boldsymbol{q} - \boldsymbol{q}_d$. Denoting estimated values due to imperfect modeling as $\hat{[\,]}$, we then define our control law as

$$\boldsymbol{\tau} = \hat{\boldsymbol{M}}\boldsymbol{a} + \hat{\boldsymbol{C}}\boldsymbol{v} + \hat{\boldsymbol{G}} - \boldsymbol{K_G}\boldsymbol{r} + \hat{\mathbf{K}}\boldsymbol{q} + \hat{\mathbf{D}}\boldsymbol{v} \tag{12}$$

where $\boldsymbol{K_G} \in \mathbb{R}^{9\times 9}$ is a positive definite diagonal matrix.

To derive a computationally efficient control law while maintaining stability, we categorize the system uncertainties into groups based on their impact and we account for them independently. We assume that the uncertainties in stiffness and damping have a higher impact in system's dynamic response than the uncertainties in $\hat{\boldsymbol{M}}$, $\hat{\boldsymbol{C}}$, and $\hat{\boldsymbol{G}}$ (AS-iv). As discussed earlier, these uncertainties are a function of pressure and deformation (AS-iii). However, we can model these deviations as an additive but variable term that is bounded with a small rate of change due to their low bandwidth [19]. Therefore, we can use adaptive terms to compensate for these uncertainties [20]. This adaptive formulation provides greater flexibility in our design, as we do not need to identify the bounds of uncertainty. Additionally, the derivation of regressor matrix for these linear terms is straightforward, and their implementation is time-efficient.

We assume that the uncertainties in $\hat{\boldsymbol{M}}$ and $\hat{\boldsymbol{C}}$ have a minimal effect on the system's dynamics (AS-i,ii), and that $\hat{\boldsymbol{G}}$ is accurate enough. Consequently, our strategy involves rejecting these uncertainties with a robust term rather than adapting to them. As discussed in Section III, we observe that deriving these terms is computationally expensive, contradicting our primary objective of designing a fast, real-time dynamic control system In particular, the regressor matrix is computationally costly. When using a robust term to reject these uncertainties, the primary challenge becomes determining the uncertainty bounds, which is typically resolved through trial and error. Given minimal uncertainty bounds on the aforementioned terms, we define the error bound as

$$\|\hat{\boldsymbol{M}}\boldsymbol{a} + \hat{\boldsymbol{C}}\boldsymbol{v} + \hat{\boldsymbol{G}}\|_i \ll K_G(i,i) \tag{13}$$

This allows us to rewrite (12) as

$$\boldsymbol{\tau} = \hat{\boldsymbol{M}}\boldsymbol{a} + \hat{\boldsymbol{C}}\boldsymbol{v} + \hat{\boldsymbol{G}} - \boldsymbol{K_G}\boldsymbol{r} + \boldsymbol{Y_p}\hat{\boldsymbol{\theta}}_{\boldsymbol{p}} \tag{14}$$

where $\boldsymbol{K_G}\boldsymbol{r}$ is the robust term, $\boldsymbol{Y_p} \in \mathbb{R}^{9\times 18}$ is the regressor, and $\hat{\boldsymbol{\theta}}_{\boldsymbol{p}} \in \mathbb{R}^{18}$ are the adaptable parameters defined as

$$\boldsymbol{Y_p} = [\mathrm{diag}(\boldsymbol{q}^n), \quad \mathrm{diag}(\boldsymbol{v})], \quad \hat{\boldsymbol{\theta}}_{\boldsymbol{p}} = \left[\hat{\boldsymbol{K}}^T, \hat{\boldsymbol{D}}^T\right]^T. \tag{15}$$

We can then define our adaptive law as

$$\dot{\hat{\boldsymbol{\theta}}}_p = -\boldsymbol{\Gamma}\boldsymbol{Y}_p^T(\boldsymbol{q}, \dot{\boldsymbol{q}}, \boldsymbol{a}, \boldsymbol{v})r, \quad \boldsymbol{\Gamma} = \begin{bmatrix} \boldsymbol{\Gamma_K} & \mathbf{0}_{9\times 9} \\ \mathbf{0}_{9\times 9} & \boldsymbol{\Gamma_D}, \end{bmatrix} \tag{16}$$

where $\boldsymbol{\Gamma_K} \in \mathbb{R}^{9\times 9}$ and $\boldsymbol{\Gamma_D} \in \mathbb{R}^{9\times 9}$ are positive definitive diagonal matrices that work as tuning gains for adaptive term for stiffness and damping coefficients, respectively. The stability of our proposed adaptive passivity control law (14), (15), and (16) was proven in [13].

C. Task Space Control

To address the real-world manipulation and movement tasks that our robot would perform, it is essential to extend our control formulation to the task space. Unlike actuator space control, where an actuator trajectory is pre-defined, task space control must generate the auxiliary actuator signals in (11) from task space information (robot's current and desired pose, velocity and acceleration profiles). We define these reference signals $(\dot{\boldsymbol{q}}_r, \ddot{\boldsymbol{q}}_r)$ as

$$\boldsymbol{v} = \dot{\boldsymbol{q}}_r, \quad \boldsymbol{a} = \ddot{\boldsymbol{q}}_r, \quad \boldsymbol{r} = \dot{\boldsymbol{q}} - \boldsymbol{v}, \tag{17}$$

Our goal is to define these reference signals to satisfy task space objectives while ensuring the solution remains within the feasible range of the actuator space, because each pneumatic actuator can only elongate within limits based on the maximum pressure it can maintain.

1) Task Space Control for Position or Orientation: We start by considering position or orientation control, introducing generalized reference signals in task space as $\dot{\boldsymbol{z}}_r$ and $\ddot{\boldsymbol{z}}_r$. Using differential kinematics, these reference signals can be translated from task space to actuator space as

$$\dot{\boldsymbol{q}}_r = ({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}})^{\dagger}\dot{\boldsymbol{z}}_r, \quad \ddot{\boldsymbol{q}}_r = ({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}})^{\dagger}\ddot{\boldsymbol{z}}_r + ({}^{\mathcal{B}}\dot{\mathbf{J}}^{\mathcal{Z}})\dot{\boldsymbol{z}}_r. \tag{18}$$

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Given the redundancy in our robot, it is necessary to use the Jacobian's pseudo-inverse $({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}})^{\dagger}$. Here, $({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}})$ indicates that the Jacobian is expressed with respect to the base frame $\mathcal{B}$ and is calculated based on the task space variable $\mathcal{Z}$.

As previously discussed, simply finding a solution is insufficient; we must also account for the system's actuator-limit constraints. To ensure actuator limit avoidance, we employ a redundancy resolution approach by leveraging the null-space projection of the redundant manipulator. This formulation at both the velocity and acceleration levels, further described in [21], [22], is given by

$$\begin{aligned}\dot{\boldsymbol{q}}_r &= \left({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}}\right)^{\dagger}\dot{\boldsymbol{z}}_r + \left(\mathbf{I}_9 - \left({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}}\right)^{\dagger}\left({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}}\right)\right)\boldsymbol{\kappa}\\ \ddot{\boldsymbol{q}}_r &= \left({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}}\right)^{\dagger}\left(\ddot{\boldsymbol{z}}_r - ({}^{\mathcal{B}}\dot{\mathbf{J}}^{\mathcal{Z}})\dot{\boldsymbol{q}}_r\right)\\ &\quad + \left(\mathbf{I}_9 - ({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}})^{\dagger}\left({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}}\right)\right)\left[\dot{\boldsymbol{\kappa}} + ({}^{\mathcal{B}}\dot{\mathbf{J}}^{\mathcal{Z}})^{T}\left({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}}\right)^{\dagger T}(\dot{\boldsymbol{q}}_r - \boldsymbol{\kappa})\right]\end{aligned} \tag{19}$$

where $\boldsymbol{\kappa}$ is designed to optimize a cost function $\boldsymbol{H}$, with $\boldsymbol{\kappa} = \nabla_{\boldsymbol{q}}\boldsymbol{H}$. In this study, we define $\boldsymbol{H}$ as

$$\boldsymbol{H} = \gamma\left(\boldsymbol{H}_1 + \boldsymbol{H}_2\right), \tag{20}$$

where γ is a tuning parameter. Since both $\boldsymbol{H}_1$ and $\boldsymbol{H}_2$ are positive cost functions, a positive γ will maximize the cost function, whereas a negative γ will minimize it. Here, we define γ to be negative.

We define $\boldsymbol{H}_1$ so the actuator variables remain within their respective lower and upper bounds, $q_{\min}$ and $q_{\max}$ [23], as

$$\boldsymbol{H}_1(q) = \gamma_1 \sum_{i=1}^{9} \frac{(q^{\max} - q^{\min})^2}{(q^{\max} - q(i))(q(i) - q^{\min})} \tag{21}$$

where γ_1 is a positive tuning variable.

For $\boldsymbol{H}_2$, we design an objective function to promote uniform deformation across the robot's sections inspired by [24]. This is achieved by minimizing the squared differences in elongation between consecutive sections as

$$\boldsymbol{H}_2(q) = \gamma_2\left[\left(\|\boldsymbol{q}_1 - \boldsymbol{q}_2\|\right)^2 + \left(\|\boldsymbol{q}_2 - \boldsymbol{q}_3\|\right)^2\right], \tag{22}$$

where γ_2 is a positive tuning parameter.

Finally, we define the reference signals. For position control, the signals are given by

$$\boldsymbol{e}_x = \boldsymbol{x} - \boldsymbol{x}_d, \quad \dot{\boldsymbol{x}}_r = \dot{\boldsymbol{x}}_d - \boldsymbol{\alpha}_x\boldsymbol{e}_x, \quad \ddot{\boldsymbol{x}}_r = \ddot{\boldsymbol{x}}_d - \boldsymbol{\alpha}_x\dot{\boldsymbol{e}}_x \tag{23}$$

where $\boldsymbol{\alpha}_x \in \mathbb{R}^{3\times 3}$ is a diagonal, positive-definite matrix. We then set $\dot{\boldsymbol{z}}_r = \dot{\boldsymbol{x}}_r$ and $\ddot{\boldsymbol{z}}_r = \ddot{\boldsymbol{x}}_r$, and substitute $({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}})$ with $({}^{\mathcal{B}}\mathbf{J}^{v})$ in (19).

For orientation control, we calculate the orientation error using the quaternion formulation from [25]. Let $\mathcal{Q}$ represent the quaternion of the current rotation matrix ($\mathcal{R} \to \mathcal{Q} = \{\eta, \boldsymbol{\epsilon}\}$), where η is the scalar and $\boldsymbol{\epsilon}$ the vector part. Similarly, the desired quaternion is $\mathcal{Q}_d$ ($\mathcal{R}_d \to \mathcal{Q}_d = \{\eta_d, \boldsymbol{\epsilon}_d\}$). orientation error is given by

$$\boldsymbol{e}_o = \Delta\boldsymbol{\epsilon} = \eta_d\boldsymbol{\epsilon} - \eta\boldsymbol{\epsilon}_d - \widehat{\boldsymbol{\epsilon}}\boldsymbol{\epsilon}_d \tag{24}$$

The derivative of $\boldsymbol{e}_o$, corresponding to the angular velocity error in the base frame, is defined as

$$\dot{\boldsymbol{e}}_o = (\boldsymbol{\omega})_{\mathcal{B}} - (\boldsymbol{\omega}_d)_{\mathcal{B}} = {}^{\mathcal{B}}\mathbf{J}^{\omega}\dot{\boldsymbol{q}} - \left(\dot{\mathcal{R}}_d\mathcal{R}_d^T\right)^{\vee} \tag{25}$$

Therefore, we can calculate the signals for orientation control, analogous to (23) as

$$\dot{\mathcal{O}}_r = (\boldsymbol{\omega}_d)_{\mathcal{B}} - \alpha_o\boldsymbol{e}_o, \quad \ddot{\mathcal{O}}_r = (\dot{\boldsymbol{\omega}}_d)_{\mathcal{B}} - \alpha_o\dot{\boldsymbol{e}}_o \tag{26}$$

where $\boldsymbol{\alpha}_o \in \mathbb{R}^{3\times 3}$ is a diagonal, positive-definite matrix. We then set $\dot{\boldsymbol{z}}_r = \dot{\boldsymbol{\mathcal{O}}}_r$ and $\ddot{\boldsymbol{z}}_r = \ddot{\boldsymbol{\mathcal{O}}}_r$, and substitute $({}^{\mathcal{B}}\mathbf{J}^{\mathcal{Z}})$ with $({}^{\mathcal{B}}\mathbf{J}^{\omega})$ in (19).

2) Task Space Simultaneous Position and Orientation Control: Despite having 9 DoF, our soft robot does not feature a wrist joint for decoupled orientation control like its rigid counterparts. This means that it does not operate within a dexterous workspace and cannot guarantee independent and instantaneous position and orientation adjustments with slight changes in actuator variables, as implied by resolved rate control. To ensure stable and satisfactory task space tracking performance, we propose a prioritization hierarchy that emphasizes position control first, followed by orientation control, and finally actuator limit avoidance and homogeneous deformation. This ensures that the robot prioritizes its primary task while avoiding configurations that would violate constraints or lead to instability. Similar to the previous section, we use redundancy resolution [16], [26] to exploit the null space of the primary task and enable the execution of secondary tasks. This allows us to compute a resolved rate solution for both velocity and acceleration levels as

$$\begin{aligned}\dot{\boldsymbol{q}}_r &= \left({}^{\mathcal{B}}\mathbf{J}^{v}\right)^{\dagger}\dot{\boldsymbol{x}}_r + \left(\overline{{}^{\mathcal{B}}\mathbf{J}^{\omega}}\right)^{\dagger}\left(\dot{\boldsymbol{O}}_r - {}^{\mathcal{B}}\mathbf{J}^{\omega}\left({}^{\mathcal{B}}\mathbf{J}^{v}\right)^{\dagger}\dot{\boldsymbol{x}}_r\right) + \mathcal{N}_2\boldsymbol{\kappa},\\ \ddot{\boldsymbol{q}}_r &= \left({}^{\mathcal{B}}\mathbf{J}^{v}\right)^{\dagger}\left(\ddot{\boldsymbol{x}}_r - ({}^{\mathcal{B}}\dot{\mathbf{J}}^{v})\dot{\boldsymbol{q}}_r\right) + \left(\overline{{}^{\mathcal{B}}\mathbf{J}^{\omega}}\right)^{\dagger}\left(\ddot{\mathcal{O}}_r - ({}^{\mathcal{B}}\dot{\mathbf{J}}^{\omega})\dot{\boldsymbol{q}}_r\right.\\ &\quad \left. - {}^{\mathcal{B}}\mathbf{J}^{\omega}({}^{\mathcal{B}}\mathbf{J}^{v})^{\dagger}\left(\ddot{\boldsymbol{x}}_r - ({}^{\mathcal{B}}\dot{\mathbf{J}}^{v})\dot{\boldsymbol{q}}_r\right)\right) + \mathcal{N}_2\boldsymbol{\kappa},\\ \overline{{}^{\mathcal{B}}\mathbf{J}^{\omega}} &= {}^{\mathcal{B}}\mathbf{J}^{\omega}\mathcal{N}_1, \quad \mathcal{N}_1 = \mathbf{I}_9 - \left({}^{\mathcal{B}}\mathbf{J}^{v}\right)^{\dagger}{}^{\mathcal{B}}\mathbf{J}^{v},\\ \mathcal{N}_2 &= \mathcal{N}_1\left(\mathbf{I}_9 - \left({}^{\mathcal{B}}\mathbf{J}^{\omega}\right)^{\dagger}{}^{\mathcal{B}}\mathbf{J}^{\omega}\right)\end{aligned} \tag{27}$$

V. Experimental Results

To assess the effectiveness of our control method, we conducted extensive experiments on the robot under varying conditions, including different speeds, loading scenarios, disturbance rejection, and trajectory complexities. Our analysis begins with system identification, followed by a detailed evaluation of actuator space control, which is then extended to task space control. We introduce our adaptive passivity controller (AP) and compare its performance against the fixed passivity controller (PA), where $\boldsymbol{\Gamma} = 0_{18\times 18}$ in all scenarios. Additionally, in joint space, we benchmark our approach against PD + Feedback Linearization (PDFL)[7]. For task space position control, we compare our method with the approach proposed by Fischer et al. [9]. Additionally, we extend their formulation to include orientation control and compare it with our method. Since our work is the first to propose prioritizing position over orientation in dynamic control, and no directly related methods were found in the literature, we compare our approach solely with the PA controller. This comparison highlights the advantages of our adaptive approach in addressing the challenges of soft robot control. The control parameters for each scenario are detailed in Table I.

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TABLE I
CONTROL PARAMETERS FOR VARIOUS CONTROL SCHEMES

Parameter	Actuator Space	Position	Orientation	Position & Orientation
$\mathbf{\Lambda}$	$8\,\mathbf{I}_{9\times9}$	—	—	—
$\boldsymbol{\alpha}_x$	—	$12\,\mathbf{I}_{3\times3}$	—	$10\,\mathbf{I}_{3\times3}$
$\boldsymbol{\alpha}_o$	—	—	$20\,\mathbf{I}_{3\times3}$	$20\,\mathbf{I}_{2\times2}$
$\boldsymbol{K}_G$	$18\,\mathbf{I}_{9\times9}$	$12\,\mathbf{I}_{9\times9}$	$9\,\mathbf{I}_{9\times9}$	$8\,\mathbf{I}_{9\times9}$
$\mathbf{\Gamma}_K$	$10^5\,\mathbf{I}_{9\times9}$	$10^5\,\mathbf{I}_{9\times9}$	$10^5\,\mathbf{I}_{9\times9}$	$10^4\,\mathbf{I}_{9\times9}$
$\mathbf{\Gamma}_D$	$\mathbf{I}_{9\times9}$	$0.1\,\mathbf{I}_{9\times9}$	$100\,\mathbf{I}_{9\times9}$	$0.1\,\mathbf{I}_{9\times9}$
γ	—	-0.001	-0.001	-0.1
γ_1	—	1	1	1
γ_2	—	0	1.5	0.1
$q^{\max}$	—	0.1	0.1	0.06
$q^{\min}$	—	0	-0.02	-0.01

TABLE II
L_2 ERRORS STATISTICS [MM] FOR JOINT SPACE CONTROL

Approach	1 rad/s	3 rad/s	3 rad/s + Dist.	6 rad/s
AP (ours)	11.86 ± 25.07	19.42 ± 21.35	22.10 ± 20.79	28.02 ± 19.72
PA	35.70 ± 19.33	35.45 ± 18.10	35.52 ± 18.13	36.67 ± 17.82
PDFL [7]	28.27 ± 20.87	38.89 ± 17.09	38.72 ± 17.09	38.63 ± 17.72

A. System Identification

Using a constrained nonlinear optimization algorithm with a cost function based on the L_2 norm between measured and model-predicted outputs for a single section, we identified the physical parameters m_i, k_i, and d_i by activating one actuator per section [13] with a chirp signal. Although uniform stiffness and damping were assumed across sections, variability in actuator properties is acknowledged. The identification was performed sequentially, starting with the first section and progressively incorporating the second and third sections to isolate the effects of the lower sections. The identified parameters for the first, second, and third sections are as follows: stiffnesses of 538, 500, and 450 N/m; damping coefficients of 130.42, 115, and 110 Ns/m; masses of 1.17, 0.54, and 0.265 kg; and actuator offsets of 46.32, 32, and 24 mm from the center, respectively. The initial length of each section is 150 mm.

B. Actuator Space Control

First, we evaluate our actuator space control strategy under the assumption that a higher-level controller generated the actuator variables from the desired pose. The desired trajectory for actuator j in section i was defined as

$$(l_{i,j})_d = \boldsymbol{q}_d(3i+j) = B_i + 0.01\sin\left(ft + (j-1)\pi/2\right) \tag{28}$$

where B_i represents the baseline elongation for section i, with values of 0.045, 0.05, and 0.04 for sections 1, 2, and 3, respectively. The controllers are implemented in four different scenarios. In the first three experiments, control performance is evaluated at frequencies of $f = \{1, 3, 6\}$ rad/s over a duration of 30 seconds per test. Additionally, to assess the controller's stability while tracking a trajectory at $f = 3$ rad/s, we introduced disturbances by forcefully hitting and poking the robot. Each disturbance test was also conducted for 30 seconds. The L_2 error statistics are represented in Table II.

AP consistently outperforms the comparative methods by effectively compensating for uncertainties, though its performance declines at higher speeds. PA maintains consistent performance across all scenarios, generally outperforming PDFL except at lower speeds.

C. Task Space Control

Next, we evaluate task-space trajectory tracking of our AP approach compared to PA and the method in [9], focusing on position and orientation tracking individually, and on simultaneous control for AP and PA. We assess system stability under varying loads, disturbance for different shapes and speeds. We attached a 200 g gripper to the robot's end plate, but this is unknown to the controller. Each trial begins with actuator-space control, moving the robot to a predefined home position over 2.5 seconds, followed by task-space control until $t = 35$ seconds, unless stated otherwise.

1) Position Tracking: To evaluate the controller's performance, we first define a circular trajectory:

$$\boldsymbol{x}_d = [0.1\sin(ft), 0.1\cos(ft), 0.6]^T \tag{29}$$

with frequencies $f = \{0.5, 1, 2, 3\}$ rad/s in a no-load condition. Next, we implement a dynamic load scenario where the gripper manipulates objects weighing between 100g and 300g. The robot follows 1.5 revolutions of the circular trajectory at $f = 1$ rad/s before suddenly releasing the object, then picks up another object within 0.5 revolutions. This cycle continues for $t = 60$ seconds. We then evaluate tracking of two complex trajectories: a 2-petal and a 4-petal shape. The 2-petal trajectory is defined as

$$\boldsymbol{x}_d = [0.1\sin(ft), 0.1\cos(ft/2), 0.6]^T \tag{30}$$

with $f = \{1, 2\}$, while the 4-petal trajectory follows

$$\boldsymbol{x}_d = [0.1\cos(ft/2)\cos(ft/4), 0.1\cos(ft/2)\sin(ft/4), 0.6]^T \tag{31}$$

with $f = \{0.5, 1, 2\}$. Finally, we test the robot's disturbance rejection by commanding it to follow the circular trajectory (29) at $f = 1$ rad/s while applying external perturbations by poking and hitting it with a wooden stick. Trajectory tracking results for selected scenarios are shown in Fig. 3, and the statistical results are summarized in Table III. As demonstrated in all scenarios, our method (AP) outperforms the comparative approaches. A clear trend is that accuracy decreases for all methods as the speed increases, yet both PA and AP remain stable, attributable to their passivity-based formulation. In contrast, the method proposed in [9] becomes unstable in two fast scenarios: circle tracking at $f = 3$ rad/s and the 2-petal path at $f = 2$ rad/s.

2) Orientation Control: We evaluated the orientation control method under both constant and variable orientations around the y-axis: (i) $\theta_y = \pi/4$, and (ii) $\theta_y = \frac{\pi}{2}\sin(t)$.

In the third scenario, the robot was tasked with following trajectory (ii) while being physically disturbed with a stick. In the fourth scenario, the algorithm was run for 60 seconds, during which the robot followed trajectory (ii) while interacting with an external load. Specifically, when the robot reached $\pi/2$, it was given multiple objects with a variety of masses of 100 g to 300 g. The robot then carried the object and released it at $-\pi/2$. The tracking performance for scenarios (i) and (ii) is shown in Fig. 4, while the error statistics for all tested methods are summarized in Table IV.

As seen in the set-point tracking results, where the robot is required to reach a target orientation instantaneously, the AP method exhibits the least oscillation, whereas both PA and Fischer [11] display oscillatory behavior. In all scenarios, AP consistently outperforms the comparative methods, highlighting

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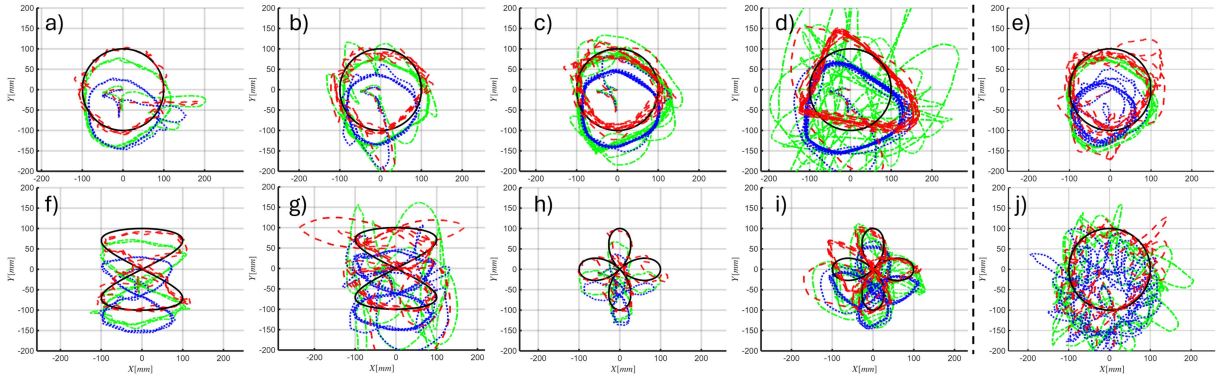


Fig. 3. Tracking performance of different controllers for various trajectories and conditions. (a)-(d) show tracking of a circle (29) at $f = 0.5, 1, 2, 3$ rad/s. (e) and (j) with the same trajectory at $f = 1$ rad/s with dynamic loading and disturbance rejection, respectively. (f) and (g) track (30) for $f = 1, 2$. (h) and (i) follow (31) at $f = 0.5, 2$. Black represents the reference, red is AP, blue is PA, and green is Fischer [9].

TABLE III
POSITION TRACKING L_2 ERROR STATISTICS [MM] FOR VARIOUS TRAJECTORIES AND CONDITIONS

Approach	Circle $f = 0.5$	Circle $f = 1$	Circle $f = 2$	Circle $f = 3$	DynamicLoad $f = 1$	2petals $f = 1$	2petals $f = 2$	4petals $f = 0.5$	4petals $f = 1$	4petals $f = 2$	Disturbance $f = 1$
AP (ours)	11.38 ± 16.08	17.58 ± 16.97	20.08 ± 14.40	43.08 ± 27.18	26.05 ± 16.47	14.27 ± 10.76	31.64 ± 30.63	10.66 ± 9.35	15.21 ± 16.91	20.28 ± 14.46	25.76 ± 24.11
PA	65.89 ± 15.83	55.13 ± 13.59	55.87 ± 8.22	62.34 ± 18.17	66.98 ± 19.09	67.15 ± 8.36	61.98 ± 15.42	56.95 ± 11.60	57.69 ± 11.66	52.07 ± 9.68	83.53 ± 27.20
Fischer [9]	49.81 ± 13.02	48.82 ± 13.84	51.86 ± 15.82	119.99 ± 58.32	51.17 ± 9.89	47.90 ± 6.04	89.82 ± 32.17	46.86 ± 5.88	49.66 ± 12.98	51.89 ± 13.78	71.65 ± 25.33

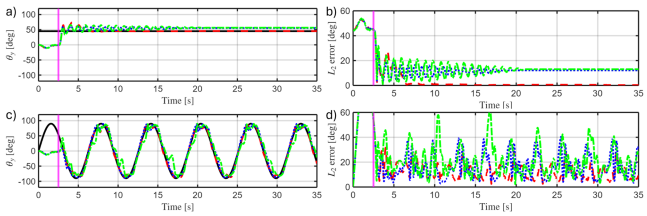


Fig. 4. Tracking performance and corresponding L_2 errors. (a) Tracking of a constant reference, $\theta_y = \frac{\pi}{4}$. (b) The associated L_2 error for (a). (c) Tracking of a sinusoidal reference, $\theta_y = \frac{\pi}{2}\sin(t)$. (d) The associated L_2 error for (c). The solid line indicates the reference trajectory, with AP shown in red, PA in blue, and Fischer et al. method in green.

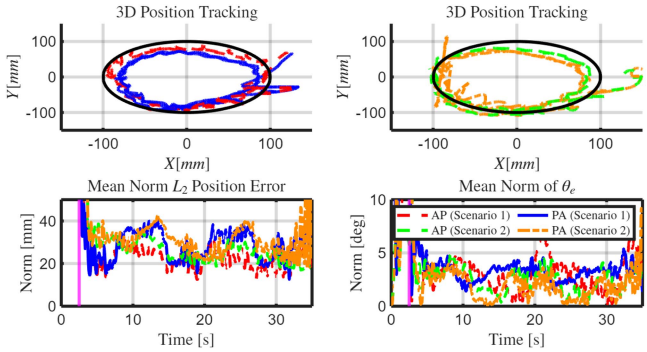


Fig. 5. Position and orientation control of a robot tracking a reference circular trajectory (solid black line) with constant orientation. *Scenario 1* has an orientation of $\theta_x = -\frac{\pi}{4}$, while *Scenario 2* has no rotation relative to the base frame.

TABLE IV
ORIENTATION L_2 ERROR STATISTICS [DEG]

Approach	$\theta_y = \pi/4$	$\theta_y = (\pi/2)sin(t)$	$\theta_y = (\pi/2)sin(t)$ Dist.	$\theta_y = (\pi/2)sin(t)$ Load
AP (ours)	2.44 ± 6.21	7.87 ± 5.40	10.35 ± 6.65	13.40 ± 8.42
PA	12.02 ± 3.84	9.83 ± 3.54	14.97 ± 6.06	14.05 ± 4.35
Fischer [9]	13.02 ± 4.77	12.22 ± 4.77	16.69 ± 6.95	16.76 ± 5.79

the importance of adaptive mechanisms in improving tracking performance and accuracy.

3) Simultaneous Position and Orientation Control: Finally, we evaluated the proposed control scheme's ability to track a trajectory while maintaining a constant orientation relative to the base frame. Without a wrist joint, simultaneous position and orientation control requires significant shape changes for twist motions. This is demonstrated computationally by the high condition number of the angular Jacobian matrix, $\boldsymbol{J}^{\omega}$, found in (27). This, combined with the control speed and numerical differentiation, complicates experimental implementation. Controlling all three orientation axes is feasible in simulation, but the oscillations it causes in experiments lead us to restrict control to the X and Y axes.

The control scheme was tested in two scenarios: (i) maintaining a constant rotation of $\pi/4$ radians around the X axis and (ii) maintaining zero rotation relative to the base frame. Tracking a circular trajectory defined as $\boldsymbol{x}_d = [0.1\sin(t/2), 0.1\cos(t/2), 0.52]$ was used in both scenarios. As shown in Fig. 5, the AP controller consistently outperformed the PA, demonstrating the value of adaptiveness.

VI. CONCLUSION

This letter addressed dynamic task space control of a redundant, pneumatically actuated soft robotic arm. To achieve real-time, numerically stable performance, we employed recursive formulations and Taylor series approximations in our dynamics calculations. By analyzing the impact of each dynamic parameter, we simplified the model by compensating for omitted terms within our control law. To handle unmodeled dynamics, external disturbances, and parametric uncertainties, particularly in stiffness and damping coefficients, we developed an adaptive passivity (AP) control method. Adaptive terms addressed the dominant dynamic effects stiffness and damping, while a robust

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control term mitigated remaining uncertainties. This approach enabled real-time control at a frequency of 1 kHz. To keep control actions within the limits of the actuators, we used a redundancy resolution algorithm and introduced a cost function promoting homogeneous deformation across all sections, enhancing motion accuracy and smoothness. Recognizing the challenge of simultaneous position and orientation control, especially without a wrist joint, we decoupled these controls using the redundancy resolution algorithm, prioritizing position control over orientation, and then assigning actuator limit avoidance and shape homogeneity as lower-priority tasks.

Experimental results showed that our adaptive control along with the redundancy resolution algorithm, outperformed the comparative methods under various challenging conditions. Specifically, our approach achieved joint space trajectory tracking at speeds of up to 6 rad/s. In addition, position control tracking attained translational speeds of 0.42 m/s and accelerations of 0.12 m/s^2. Furthermore, orientation control reached 1.57 rad/s, and simultaneous position and orientation control achieved speeds of 0.07 m/s. To our knowledge, this is the first successful implementation of dynamic task-space control for a redundant soft robot, managing both position and orientation while avoiding actuator limits on real hardware. This has enabled us to achieve higher speeds compared to the state of the art. While omitting Coriolis effects and certain minor dynamic terms is justified for our current operating speeds, we acknowledge that more agile tasks may require advanced compensation techniques or data-driven approaches.

Future work will focus on improving the arm's agility by leveraging system dynamics through optimization-based control and trajectory optimization techniques. More importantly, we will explore methods to achieve agile, full 3D position and orientation control. Furthermore, because our method relies on a recursive dynamic formulation, it can be readily extended to higher DoF soft robots, thus offering a scalable framework for more complex systems. By advancing control methods for soft robots, the proposed work has the potential to contribute to improvements in speed and precision for applications such as agricultural harvesting [27], [28] and surgical interventions [29], where robust and accurate performance is essential. Our adaptive dynamic control framework offers a promising direction for enhancing the versatility of soft robotic arms across diverse real-world applications.

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